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Plant functional groups modulate variation and covariation in leaf and seed traits across years

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Abstract

Leaf and seed traits are key determinants of plant growth and reproduction, mediating how plants respond to environmental change. Leaf and seed traits can also covary. For example, species with larger seeds tend to have larger leaves. However, the covariation between leaf and seed traits has not been fully resolved due to the confounding effects of plant functional groups and climate variability. Using a sample of 33 common herbs in Songnen grassland, we explored whether seed traits were associated with leaf traits, and whether the correlations between leaf and seed traits were constrained by life history (or growth form) and climate change. We found significant interspecific trait variation ($\geq 95\%$) and functional group differences in most traits. Annual species and forbs exhibited strong positive correlations between leaf and seed traits (e.g., leaf number vs. seed number), while relationships were decoupled for perennial grasses. Unexpectedly, the relationships between leaf and seed traits differed between sampling years. For species in 2017, seeding intensity increased with leafing intensity; while, in 2016, this positive trend was not significant. The relationships between leaf and seed traits are influenced by functional groups, and affected by interannual climate variability mediated by changes in plant individual size. Our results highlight species-specific traits and their correlations in response to environmental conditions, providing critical insights for predicting grassland community responses to climate change.

Keywords climate change, extreme rainfall events, plant functional groups, seed-leaf correlations, seeding intensity

Introduction

Plant traits and their covariation vary dramatically among species and functional groups, which is instrumental for understanding how plant functional and life history strategies are constrained by evolution and the environment [1–3]. Variation in trait correlations for different plant species or functional groups has been demonstrated extensively [4–6]. For example, annual species are likely to possess ‘fast’ economic traits, which are characterized by high leaf area, high leaf nitrogen content per leaf mass and short leaf lifespan. But perennial species exhibit ‘slow’ traits, characterized by high leaf dry matter content and low leaf mass per area [4, 7, 8]. Covariation of multiple traits associated to a multidimensional space

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provides insight into the adaptive strategies employed by plants that may permit species to become common or dominant in communities with various environments [9–12]. However, multidimensional trait analyses rarely connect leaves and seeds which represent two of the most important life history processes of plants: growth and reproduction.

The Leaf-Height-Seed (LHS) scheme is widely cited as expressing important strategic variation between species based on their location in a three-dimensional space defined by three functional traits: specific leaf area, plant height, and seed mass [13, 14]. For the size relationship between seeds and leaves predicted by seed-phytomer-leaf (SPL) theory, larger seeds sprout larger seedlings, which in turn can support larger leaves [15]. According to the predictions of the ‘leafing intensity premium’ hypothesis, there is a significant correlation between the number of leaves and seeds per unit of stem volume. In other words, the greater the leaf number per stem volume, the greater the fruit number per stem volume resulting in more seeds produced [16, 17]. The positive relationship of seed mass and leaf mass, as a result of physiological and metabolic constraints, has been reported in several studies [18, 19]. However, considerable uncertainty remains regarding the relationship between leaf and seed traits. One possible reason is that different growth forms and vegetation types leads to a large range in size and number of leaf and seed variation across species, sometimes spanning several orders of magnitude, which in turn leads to more uncertainty in the relationship between leaves and seeds [20, 21]. On the other hand, leaf and seed quantitative traits are not independent of size. Leaf and seed number increase with plant above-ground biomass. Therefore, the expression and correlation of plant traits may vary due to changes in plant size, which may be influenced by the environment, such as precipitation patterns during growth. Consequently, understanding how plant functional traits respond to climate variability is crucial for comprehending plant community stability and adaptability.

It has been demonstrated that climate variability exerts a significant effect on individual plant traits, such as seed size, plant height and leaf carbon content [22, 23]. This effect is especially strong for species in arid and semi-arid grasslands, where increases in precipitation variability have been shown to shape plant functional traits, including the leaf economic traits [24–26]. Specifically, summer drought decreases leaf stomatal conductance, whereas wetter conditions increase leaf water content and stomatal conductance [26]. Regarding trait covariation, when rainfall is plentiful, a positive correlation has been observed between mass-based photosynthetic rate and leaf nitrogen concentration, and a negative correlation has been observed between mass-based photosynthetic

rate and leaf mass per area [27, 28]. When rainfall is scarce, these relationships are weak or entirely absent [29]. Further analysis has indicated that directional selection for individual traits (e.g. high mass-based photosynthetic rate and low leaf nitrogen concentration) predominated under low precipitation conditions, whereas correlational selection for trait combinations (e.g. high leaf mass per area with low leaf nitrogen concentration) emerged under high precipitation conditions [29].

Nevertheless, the existence of a link between leaf and seed traits for different functional groups remains contentious, and the respond of this link to climate variability is unclear. This knowledge gap is especially prominent when considering the potential effect of interannual changes in precipitation patterns. The range of interannual rainfall fluctuations has widened, with extreme rainfall events occurring more frequently [30, 31]. Extreme rainfall events provide a rapid and substantial water supply, stimulating increases in canopy cover, leaf area and plant height [32]. Therefore, investigating how interannual climate change (especially contrasting precipitation patterns) modulates leaf-seed trait covariation across different plant functional groups is crucial for uncovering the mechanisms underlying the divergence of plant adaptive strategies and improving the accuracy of predicting grassland ecosystem responses to future climate scenarios.

In this study, we offer a comprehensive analysis of 33 herbaceous species across two years with contrasting precipitation patterns in Songnen grasslands by focusing on the covariation of leaf and seed traits to address the following questions: (1) How do leaf and seed traits and their covariation differ across functional groups? (2) Do contrasting precipitation patterns alter leaf-seed trait relationships, and is this mediated by plant size? (3) Are trait trade-offs (e.g., seed size vs. number) consistent across functional groups and years? (4) What is the combined effect of functional groups and precipitation patterns on seed production via trait covariation?

Our study aims to fill a critical knowledge gap by exploring the interactive effects of functional groups and precipitation patterns on leaf-seed trait covariation. The findings will improve predictions of grassland community dynamics under climate change and inform conservation strategies for semi-arid ecosystems.

Materials and methods

Study site

We studied the relationship between leaf and seed traits of herbs in the Songnen grassland (44°33′ N, 123°31′ E, 150 m a.s.l.), northeast China. This region has a continental monsoon climate with warm, humid summers and cold, dry winters; the mean annual precipitation (1980–2020) is 400 mm, with a concentrated distribution

resulting in more than 80% of the rainfall occurring between June and September. As a result of soil salinization, plant communities have low species richness; the species distribution is scattered, and individual differences of the same species within different communities are large. The dominant plant species include *Leymus chinensis* (Trin. ex Bunge) Tzvelev, *Aster pekinensis* (Hance) F. H. Chen, and *Suaeda glauca* (Bunge) Bunge [33].

Precipitation data

Mean annual growing season precipitation (GSP) from 2011 to 2020 was 346.67 mm. Total precipitation was 382 mm (2016) and 378.8 mm (2017), but patterns differed drastically: 2016 had evenly distributed monthly precipitation (May: 93.8 mm, June: 124.4 mm, July: 74.6 mm, August: 89.2 mm), while 2017 was drought-prone (May: 53.2 mm, June: 15.9 mm, July: 81.6 mm, August: 228.1 mm) with extreme rainfall events (August 3: 154.3 mm) accounting for 40.7% of growing season precipitation (Figure S1).

Species sampling

Plants were located in the grazing exclusion areas to avoid the confounding effect of livestock damage. A total of 33 herbaceous species were collected at the Songnen Grassland Research Station (44°33'N, 123°31'E). We measured traits of the same 33 species in 2016 and 2017. The 33 species were classified into four functional groups based on life history (annual/perennial) and growth form (forb/grass) including 10 annual forbs, 9 perennial forbs, 5 annual grasses and 9 perennial grasses (Table S1). Functional groups are defined by the interaction of life history (annual/perennial) and growth form (forb/grass), e.g., annual forbs (annual: life history + forb: growth form), capturing combined effects of these two traits.

Using common protocols for each species in the two sampling years, 20 undamaged individuals were collected in the core experimental area of the Songnen Grassland Research Station ($\approx 10 \text{ km}^2$). To avoid resampling, Every individual plant of each species was collected from 20 randomly selected plots (plots were spaced at least 10 m apart) including as much intraspecific size variation as possible for each species. Sampling was also independent across years (different individuals sampled in 2016 and 2017) to avoid resampling effects. Specific sampling times were based on the availability of species at the end of the growing season (September 1–15). For each individual plant, we sampled after it had completed fruiting, and we did not consider plants that had begun to fruit but still had flowers to ensure that we were able to obtain all seeds. A range of plants of different sizes for each species was included to compare the effect of plant size.

Trait measurements

Each plant was separated into leaves, stems, and seeds. All experimental material was oven dried at 70 °C for 48 h. The individual plant size was defined as the total aboveground biomass (including stem mass, leaf mass and seed mass). The total number of seeds (SeedN) and leaves (LeafN) were measured directly for all 20 individuals of each species in 2016 and 2017.

To determine the stem volume (V_{stem}) for a given plant, we summed the volume of each branch (V_{branch} , mm^3). For each branch, V_{branch} was estimated by measuring the basal branch diameter (D_{branch} , mm, the mean value of two perpendicularly intersecting diameters) and length (L_{branch} , mm), assuming a cone geometry: $V_{\text{branch}} = 1/3 \cdot \pi \cdot L \cdot (r^2 + R^2 + rR)$, where l is the length of the branch, R is the radius of the branch bases and r is the radius of the branch tip [33]. Branch diameter was measured with a digital vernier caliper (precise to 0.01 mm) by averaging two perpendicular diameters.

The seed size was defined as seed mass per seed, seedling intensity was calculated as total seed number divided by stem volume ($\text{SeedN} / V_{\text{stem}}$). Leaf size and leafing intensity were calculated in the same way as for seeds.

Statistical analysis

All analyses were performed in R ver. 4.0.0. We used analysis of variance (ANOVA) to partition variance to interspecific and intraspecific sources and to assess the partitioning of interspecific trait variability (two-way ANOVA: life history $\times$ growth form and one-way ANOVA: functional groups). Most of the trait variability was between species, and our predictions focus on interspecific variability. Thus, all other analyses were conducted using mean trait values for each species measured in 2016 and 2017. Paired t-tests are used to examine trait differences among species and functional groups across different years. We used Pearson correlations to calculate interspecific correlations among seed traits and leaf traits, which revealed a significant correlation between response variables.

The standardized major axis (SMA) regression was used to analyze the relationships of seed, leaf and individual size traits at both the inter- and intra-species level, which can be expressed as $\log y = \log b + a \cdot \log x$, where a is the scaling exponent and b is the allometric coefficient, in the SMATR package [34]. When analyzing the scaling exponents at the species level, the number of plants sampled for each species was identical; therefore, no correction for sample size was required. We employed Principal Component Analysis (PCA) with the 'psych' package to condense the complexity of plant traits (including plant size, number of leaves, and seed size) into fewer dimensions. The PCA was run on the combined dataset, which including all species in 2016

and 2017 to ensure consistent rotation and loading factors for the interannual comparison. While incorporating multiple traits in PCA may reduce the explanatory power of the primary axis, the principal components distributed along this axis more effectively capture the trends in variation among traits.

Results

Traits variance among different functional groups and years

There was substantial variability in leaf and seed traits. Most variation in leaf and seed traits among individuals was between species ($\geq 95\%$) for each year. For individuals sampled in both years, $\leq 2\%$ of trait variability was partitioned to intra-species level. While more than 65% was attributed to functional group differences (Table S2). Much of the interspecific trait variability was explained by life history and growth form (Table 1). The difference between plants within different life history and growth form groupings, that is, the effect of plant functional groups, was significant ($P < 0.05$) for all traits except seed size (Table 1).

Significant interannual variations were observed in both the number and mass of seeds. However, seed size and seeding intensity were unchanged across years. Specifically, seed number and mass were significantly higher for annual plants and perennial forbs in 2017 than in 2016. The opposite trend was observed for perennial grasses. Leaf traits showed no significant interannual variation (Fig. 1).

Leaf and seed traits covariance among different functional groups and years

Although not all seed and leaf traits exhibited significant differences across years and functional groups, in general, the covariation relationships among these traits varied between different years and functional groups. For each functional group, the seed number of annual forbs showed a significant positive correlation with leaf number. Seed size and leaf size of annual grasses also exhibited a significant positive correlation, as did the leaf mass and seed mass of both annual grasses and forbs. The seeding intensity and leafing intensity of annual grasses

exhibited a significant positive correlation. However, these trait relationships were not significant for perennial forbs or grasses (Fig. 2). The small sample size of annual grasses (5 species) may limit statistical power, which could explain why some correlations were non-significant—future studies should include more grass species to validate these patterns.

Trait coordination between leaf and seed traits were inconsistent across years. Results of the standardized major axis (SMA) regression analysis indicated that seed number significantly increased with leaf number at the species level across years. In 2016, no significant relationship was evident between the leaf and seed number; in 2017, however, the relationship was significant (Fig. 3a, Table S3). A similar trend was observed for the relationship between seed and leaf mass (Fig. 3c, Table S3). The relationship between seeding intensity and leafing intensity was marginally significant in 2017, but not significant in 2016 (Fig. 3d, Table S3). At the intraspecific level, the number of species exhibiting significant seed-leaf correlations was higher among annual forbs, annual grasses and perennial forbs in 2017 than in 2016. However, perennial grasses showed the opposite trend (Table S3).

Trade offs between size and number of seeds and leaves

A negative allometric relationship with a slope of -1.33 represented the trade-off between the number of seeds and seed size. The relationship between seeding intensity and seed size was also allometric (slope = -1.20) (Figure S3b, Table S4). There was a significant relationship between leaf number and leaf size for species in 2016 and 2017. For all species, smaller leaves generally had a higher leafing intensity (Figure S3c, Table S4). At the intraspecific level, significant trade-off relationships between seed size and seed number were detected in three species in 2016 and seven species in 2017, while for seed size and seeding intensity, a trade-off was evident for nine species in 2016 and ten species in 2017 (Table S4).

Synthetic analysis of the effects of interannual climate and plant functional groups on seed and leaf traits

Principal component analysis revealed that the first principal axis (PC1) correlated with plant size, explaining

Table 1 Mean differences among plant functional groups defined by life history (annual or perennial) and growth form (graminoid or forb or shrub)

Life history	Seed number	Seed size	Seed mass	Seeding intensity	Leaf number	Leaf size	Leaf mass	Leafingintensity
	4.26*	0.22	5.64*	7.25**	8.78**	16.55***	2.28	7.37**
Growth form	0.91	2.47	16.70***	0.82	74.61***	11.61**	16.21***	14.66***
PFG	3.73*	0.15	7.71***	3.09*	28.52***	10.18***	8.77***	7.34***

For life history and growth form, F values are derived from two-way ANOVA model with separate categorical variables representing each categorical trait. For plant functional types (PFTs) defined by both life history and growth form (as in Fig. 1), F values are derived from one-way ANOVA model. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

*** $p < 0.001$
** $p < 0.01$
* $p < 0.05$

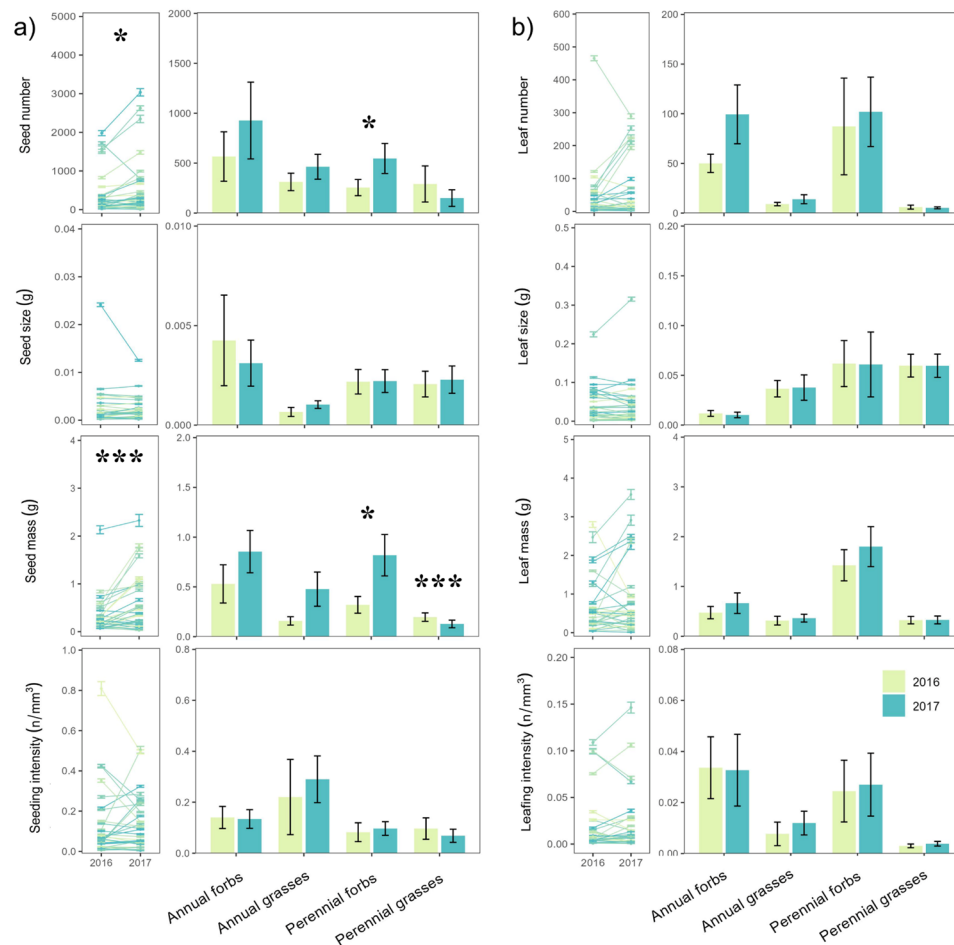


Fig. 1 Mean trait values for species separately in different plant functional groups in 2016 and 2017, respectively. Asterisks symbolise non-overlapping 95% CI between traits within each functional groups respectively. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

33.7% of the variance. Plant biomass, leaf mass, and seed mass contributed significantly to PC1. The second principal axis (PC2) accounted for 22.3% of the variance, with leaf and seed number, size, and intensity making substantial contributions (Table S5). PC2 represented relationships between seed and leaf traits (Fig. 4a). Following dimensionality reduction of traits, paired t-tests were conducted to compare differences. Significant variation was observed between years for both PC1 and PC2, indicating that interannual climate fluctuations had a significant influence on plant size and leaf-seed trait relationships. Analyses of individual functional groups revealed significant interannual differences for PC1 and PC2, for both annual and perennial forbs, demonstrating divergent responses to precipitation change across functional groups (Fig. 4b).

Our structural equation model (SEM) further demonstrates the direct and indirect effects of interannual climate variability and plant functional groups on seed number. Interannual climate variability influences seed number by affecting both aboveground plant biomass

and, subsequently, leaf biomass. Plant functional groups exert a direct negative effect on seed number and also exert an indirect effect via their influence on leaf biomass. Our SEM additionally reveals a trade-off between seed size and number - seed size exerts a direct negative effect on seed number (Fig. 5).

Discussion

As expected, significant differences were observed in both trait values and trait relationships across functional groups. The relationship between plant leaf and seed traits also varied markedly across years. Our results on annual grasses are consistent with the prediction of the leafing intensity premium theory: plants with higher leafing intensity generally had a higher seeding intensity [16]. Our annual grass dataset also supports the seed-phytomer-leaf (SPL) theory, whereby leaf size is greater when seed size is larger [15]. In the year when typical precipitation patterns occurred, the correlation between leaf and seed traits was weak or absent. However, in the year with uncharacteristic precipitation variability, this

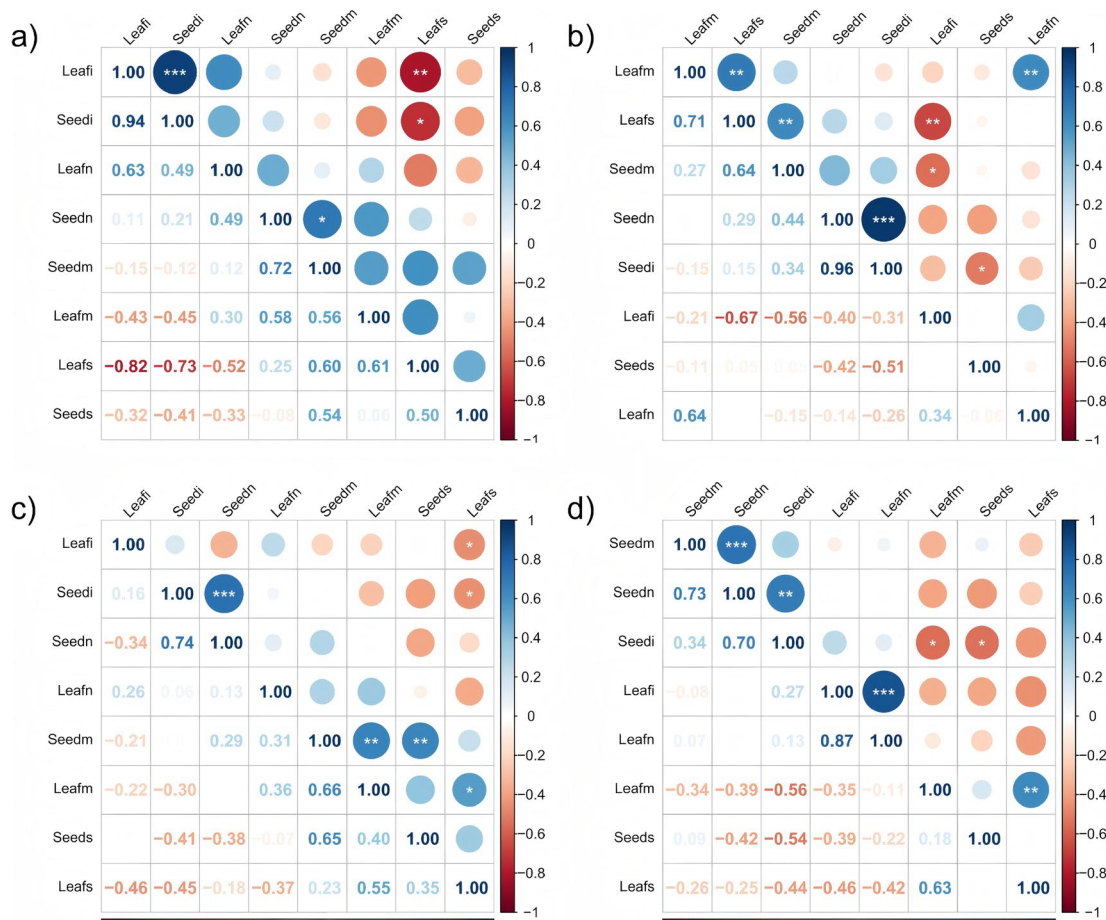


Fig. 2 Trait correlation matrix with Pearson's r coefficient across different functional groups **(a)** Annual grasses; **(b)** Perennial grasses; **(c)** Annual forbs; **(d)** Perennial forbs. Leafi: Leaf intensity; Leafs: Leaf size; Leafn: Leaf number; Leafm: Leaf mass; Seedi: Seed intensity; Seeds: Seed size; Seedn: Seed number; Seedm: Seed mass

relationship became significant, potentially mediated by differential responses among functional groups.

Seed and leaf traits variation across years within different functional groups

Most of the variation in traits of leaves and seeds was between species and functional groups, as noted in previous studies of trait variation across species [4]. Intraspecific trait variability accounted for just 1–5% of variation among individual plants (Table S2). Forbs and grasses have overlapping distributions of seed number and seedling intensity, while the forbs have higher leaf number and leafing intensity than grasses because forbs are generally have more leaves with a smaller size compared to grasses [35]. We show that herbaceous species with different life histories and growth forms have distinct phenotypic distributions of leaf and seed traits (Table 1; Fig. 1). Consequently, the plant functional groups defined by life history and growth form in our study, provide a universal view of plant trait variation and have substantial value in

describing and distinguishing plant form and function in grasslands [36–38]. Species of different functional groups exhibit specific responses to variations in precipitation. For a widely distributed annual species, intraspecific trait variation is significantly influenced by variations in precipitation. The results of our study also demonstrate that there was a significant increase in both the number and mass of perennial forbs in 2017 compared to 2016. For annual grasses and forbs, although not statistically significant, both seed number and mass were higher in 2017 than in 2016. However, for perennial grasses, both the number and mass of seeds were higher in 2016 than in 2017. This finding suggests that annual grasses and forbs exhibit greater sensitivity to extreme rainfall events, while the seed traits of perennial grasses suggest that species in this group are adapted to relatively constant precipitation patterns. The differing responses of these functional groups to precipitation variability resulted in significant trait differences across species in 2016 and 2017.

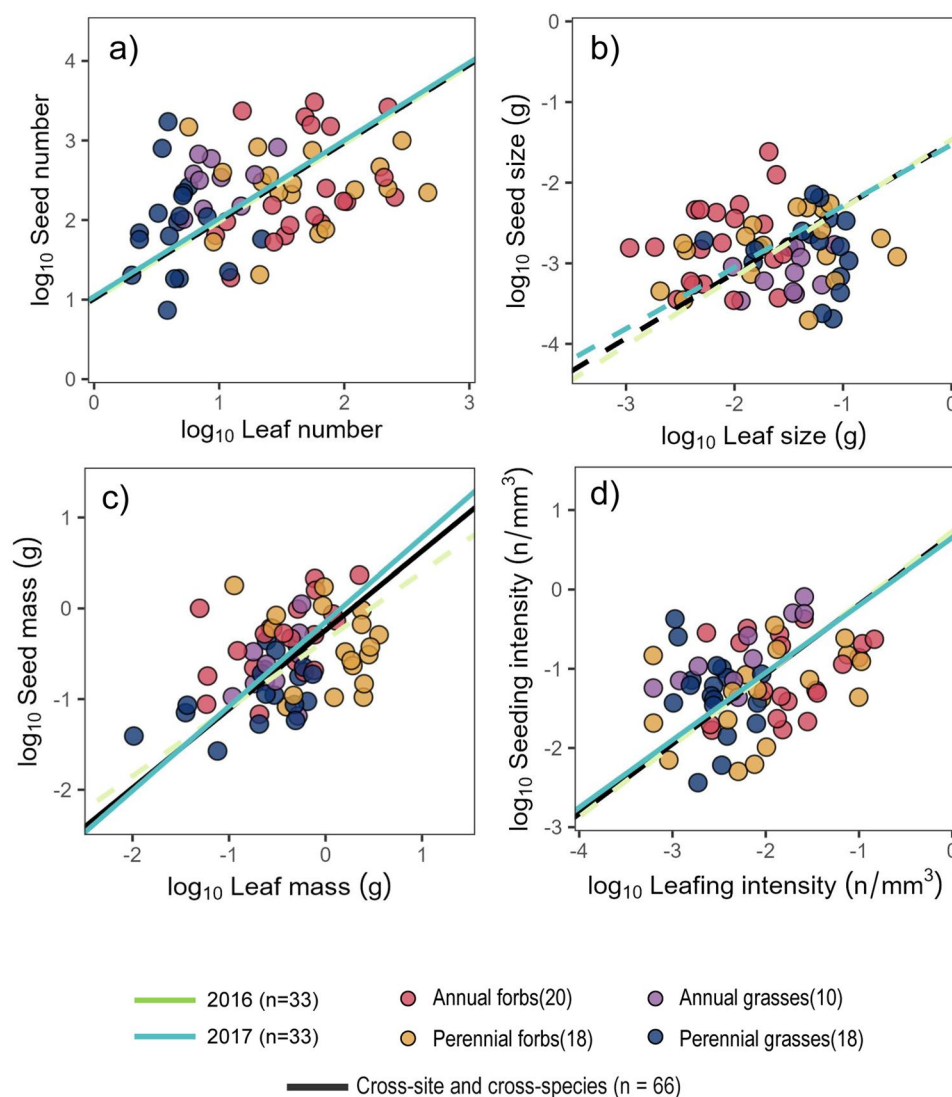


Fig. 3 Inter-species correlations between leaf and seed traits ($n=66$). Solid lines show linear relationships when $P < 0.05$ and dashed lines show linear relationships when $P > 0.05$. Different colored lines show the linear relationships for different years, circular symbols in different color represent different functional group, as indicated in the legend. See Supporting Information Tables S4 for additional statistics describing these correlations

Leaf and seed traits covariance among different functional groups

Leaf and seed traits have helped to distinguish major growth and life history strategies in plants and have demonstrated the generality of plant trait relationships [39, 40]. One such generalized relationship between traits is the tendency for large seeds to produce large seedlings. Large seedlings also tend to have large phytomers, potentially resulting in the development of stout stems with large leaves [15]. This observations is consistent with Corner's rule, which states that larger leaves and fruits occur on thicker stems, meaning that larger leaves are usually associated with larger fruits which produce larger seeds [41]. However, our results showed that the seed size of annual grasses exhibited a significant positive correlation with leaf size (Fig. 2). At the cross-species level,

no significant correlations were observed in either 2016 or 2017 (Fig. 3b). This could be because this relationship is likely to be weak in herbaceous plants with generally small seeds, the production of which is more likely to be influenced by plant life history strategies and environmental factors [9].

A positive correlation was observed between leafing intensity and seeding intensity for annual grasses (Fig. 2). Recent studies have shown that leafing intensity has a significant effect on plant reproductive biomass allocation [16, 17]. This is likely to be because some of the resources derived from photosynthesis in leaves and the carbohydrates produced by photosynthesis in the pericarp are used for seed growth [37, 42, 43]. However, for the perennial grasses of this study, the relationship between leafing and seeding intensity was decoupled.

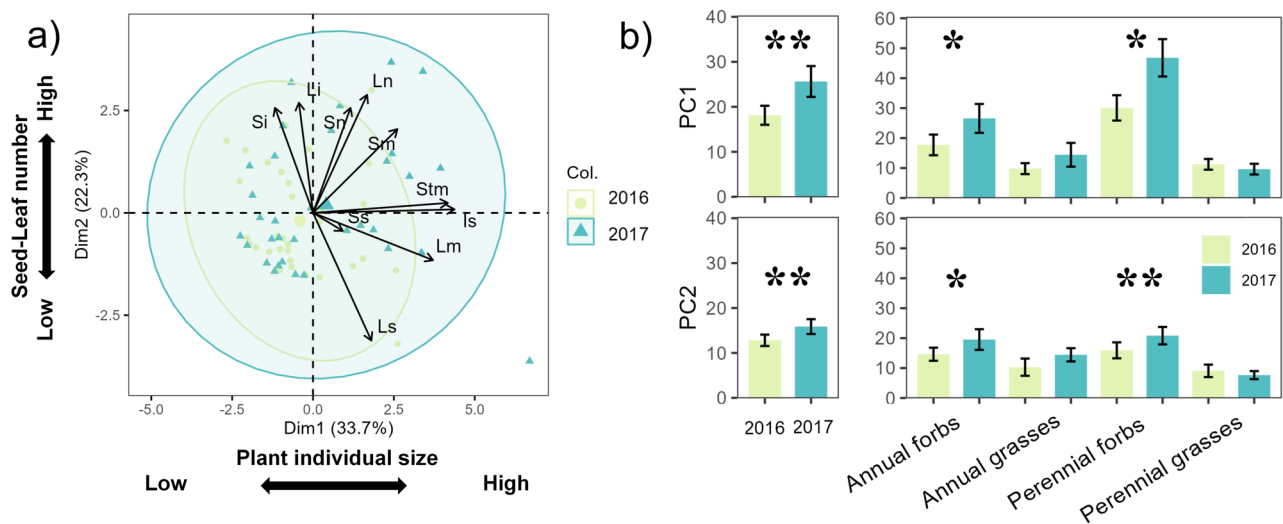


Fig. 4 Principal component analysis (PCA) of plant individual size, leaf traits and seed traits (a), and test of the distribution of traits along the space represented by the first and second dimensions of each species PCA in 2016 and 2017, respectively (b). Significant differences between treatments were tested with paired T- test (*, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$). Outputs of the PCAs are given in Supporting Information Tables S6, S7

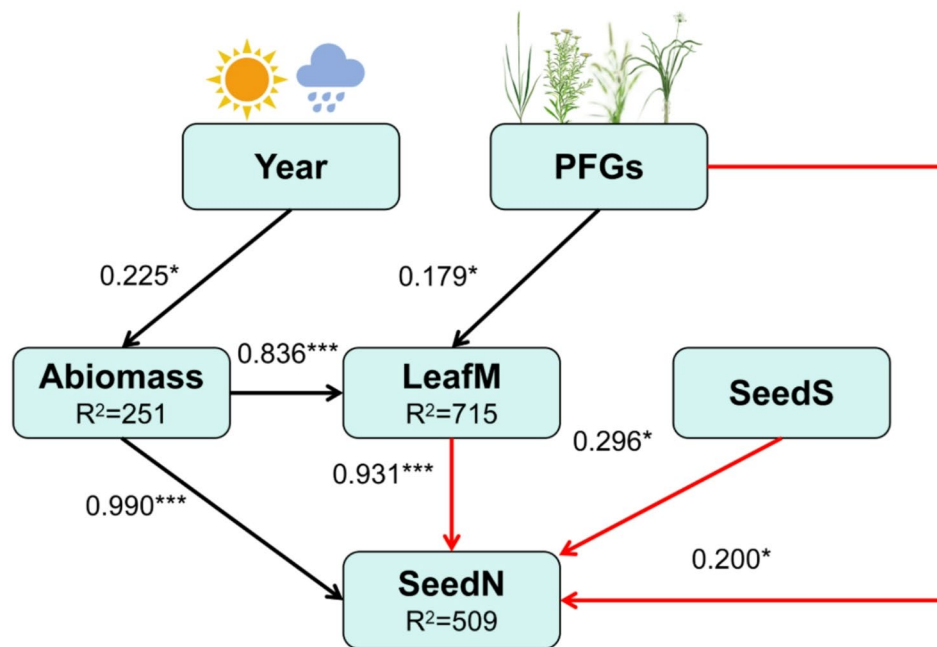


Fig. 5 Structural equation model assessing year, plant functional groups (PFGs), plant above-ground biomass (Abiomass), leaf biomass (LeafM) and seed size (SeedS) contributing to seed number variation. The numbers adjacent to arrows are path coefficients; they show the strengths of the effect. The R^2 next to the response variables indicates the proportion of variance explained. Black arrows indicate significantly positive effects, red arrows indicate significantly negative effects. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. The results of model fitting are $n=66$, $df=5$, $p=0.478$, root mean square error of approximation (RMSEA) < 0.001 , Comparative Fit Index (CFI) = 1.000

This could be because of the fundamental trade-off associated with the allocation of net mass gain into growth versus reproduction [44]. Previous research has shown that as reproductive allocation increases, the growth rate of the plant canopy will decline, as energy investment shifts from growth to reproduction [19, 45]. The trade-off between growth and reproduction is especially

important for survival in perennial species, and thus the trade-off relationship may be able to explain more of the variation in seed number and seeding intensity [46]. For the perennial grasses, the trade-off between sexual and asexual resource allocation is also an important factor affecting seed number. Most perennial grasses produce new plants through asexual reproduction. There is

a fundamental shift in resources into biomass components that facilitate sexual reproduction, which may lead to decoupling of seeds from leaves [20, 36, 37]. Another possible explanation for seed and leafing intensity to be decoupled is that the number of seeds in a perennial plant does not depend on the resources captured by the leaves in the current year, but may depend on leaves that have been retained for several years [35, 41, 47]. For perennials, the allocation of resources to the root system to ensure growth in the following year also affects the biomass allocation for reproduction in the current year [36]. Further study based on time series datasets of plant reproduction is required to resolve these uncertainties [48], but our results clearly show that trait relationships can be fundamentally different for each functional group. Our results show that leaf number and leafing intensity is negatively correlated with leaf size (Figure S3c, d). In 2016, the relationship between seed size and number was isometric, but in 2017, the trade-off was allometric. The annual species in our study increase their investment to seeds in direct proportion with plant growth [19]. In contrast, this relationship for perennial species is allometric. This may be due to the fact that annuals allocate most of their resources to reproduction, while perennials allocate a portion of their resources to storage organs [47].

Changes in leaf and seed trait covariance in different years

Interannual climate variability, particularly precipitation changes, is considered to be a critical factor affecting plant growth and reproductive traits [48]. The experimental manipulation of water availability has been used to show that the integration of key leaf economics traits is strongly dependent on precipitation, with the strongest trait covariation and highest phenotypic integration occurring under high precipitation [29]. Our findings also reveal that in years with high variability in precipitation patterns, a significant positive correlation exists across species levels between plant leaf number and seed number. Seeding intensity also exhibited a marginally significant upward trend as leafing intensity increased in 2017. However, in 2016, when rainfall was more typical, no significant correlations were found between leaf and seed traits (Fig. 3, Table S3). These correlations may represent a key mechanism by which plants respond to environmental change. For instance, grass species may adopt an opportunistic strategy in arid environments. Conversely, under similar arid conditions, forbs may operate more conservatively [49]. These contrasting responses to climatic conditions demonstrate that even within herbaceous communities, distinct functional groups employ fundamentally divergent responses. Similarly, our results also revealed that the individual size (PC1) and leaf-seed trait relationships (PC2) of annual species and forbs were significantly higher in 2017 than in 2016 (Fig. 4).

From our results, there is strong evidence that both plant size and traits correlations act synergistically to determine seed number. This suggests that seed number is affected by the limitations to plant size imposed by the inter-annual climate variations, and the leaf and seed trait combinations, which are influenced by plant size and functional groups (Fig. 5). For herbaceous and woody plants, the arithmetic basis for determining the number of seeds per plant is underpinned by the negative correlation between seed size and number and the positive correlation between plant size and seed mass [16, 17]. The often overlooked effect of individual size has resulted in an incomplete characterization of the variation in seed number in relation to environmental factors and comparisons between different life forms [19, 47]. The correlation between plant leaf and seed traits in this study is consistent with previous theoretical predictions [15, 16], but our results further indicate that the correlation between plant leaf and seed traits is constrained by plant life-history and growth form, which provides an important insight into the relationship between plant trait variation and the response of plants to environmental change.

Limitations

Unfortunately, our regionally based study of plant trait variation is limited by the number of species and years that could be practically sampled. We expect that studies or meta analyses of relationships between leaf and seed traits at a global scale will show broader trends which could further improve the theoretical basis for distinguishing plant functional groups and quantifying plant responses to the environment, as illustrated in the LHS scheme [13].

Conclusions

Our results showed that variation in, and covariation among leaf and seed traits in different years changed significantly in different functional groups. Annual species and forbs exhibit strong leaf-seed correlations, while perennial grasses show decoupled relationships. Inter annual climate variability strengthened trait covariation via increased plant size. These findings advance our understanding of plant adaptive strategies and provide a framework for predicting grassland responses to climate change. Functional group identity must be incorporated into ecosystem models to improve conservation and management outcomes, such as prioritizing perennial grasses for grassland stability conservation.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12870-026-08496-9>.

Supplementary Material 1.

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Authors' contributions

Yingxin Huang and Paul L. Drake designed the study. Changxing Fu, Heqi Wang and Gaohua Fan collected the data. Changxing Fu and Gaohua Fan analysed the data and wrote a first draft of the manuscript. Yingxin Huang and Congwen Wang contributed to further revisions. All authors read and approved the final version.

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Data availability

The species level seed and leaf traits data-set are available at ResearchGate (https://www.researchgate.net/publication/400934950_Seed_Leaf_Traits_Correlation).

Declarations

Ethics approval and consent to participate

All species were collected from the Songnen Grassland Research Station of Chinese Academy of Sciences, which is affiliated with the Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences. Consequently, the collection and utilisation of plant samples for scientific research within the research station is permissible. And all species were identified by Yingxin Huang.

Consent for publication

The authors confirm that all participants involved in this study have provided written consent for publication of the research results.

Competing interests

The authors declare no competing interests.

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